

4.4 Surface Area; Parametric Surfaces_Guide

Monday, January 12, 2026 10:42 PM

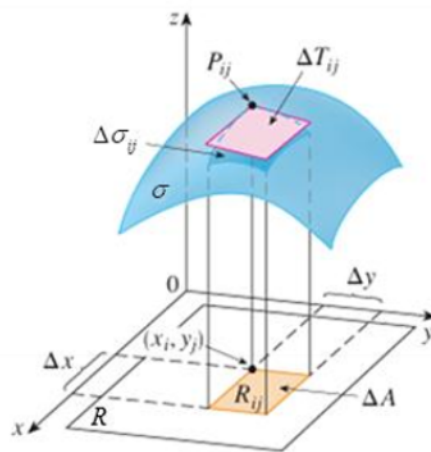


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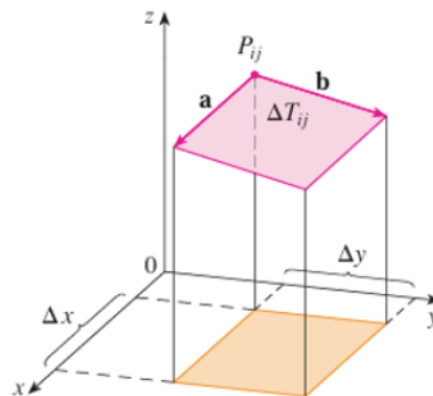
Motivation: Suppose that σ is a surface with equation $z = f(x, y)$ where f has continuous partial derivatives. Suppose also that $f(x, y) \geq 0$, and we wish to find the surface area of σ that exists over some region R in the xy -plane.

Our strategy for finding the surface area will be to divide R up into small rectangles R_{ij} with area $\Delta A = \Delta x \Delta y$ and thereby split up σ into smaller “patches” which we will call $\Delta \sigma_{ij}$. We then want to approximate the area of each $\Delta \sigma_{ij}$ over the rectangle R_{ij} . A picture of what a $\Delta \sigma_{ij}$ over R_{ij} might look like is shown at the right.



In actuality there would be many “pieces” $\Delta \sigma_{ij}$ of σ like the one pictured because there would be many rectangles R_{ij} . We are just focusing on one of the pieces.

Our plan is to now consider the point $P_{ij}(x_i, y_j)$ in R_{ij} closest to the origin, then $(x_i, y_j, f(x_i, y_j))$ is a point on the surface directly above the corner of R_{ij} closest to the origin. At this point we will construct a tangent plane (shown in pink above) and we will use the area of the tangent plane ΔT_{ij} (which is just the area of a parallelogram) to estimate the area of the surface $\Delta \sigma_{ij}$.



Now suppose we build vectors $\mathbf{a}$ and $\mathbf{b}$ that start at P_{ij} and lie along the sides of the parallelogram (as shown in the image at the right). We let ΔT_{ij} represent the area of this parallelogram. We note that as R_{ij} gets smaller and smaller ΔT_{ij} will do a better job of approximating the “piece” of σ . This leads to the following definition.

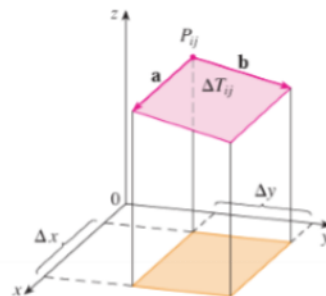
Definition: The surface area of σ (as described above) is

$$S = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n \Delta T_{ij}$$

Finding ΔT_{ij} (the area of the portion of the tangent plane we desire) can be done easily using ideas from Chapter 1. Recall that in order to find the area of a parallelogram determined by vectors $\mathbf{a}$ and $\mathbf{b}$ we simply compute

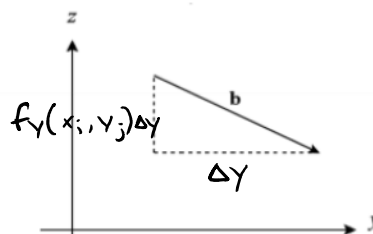
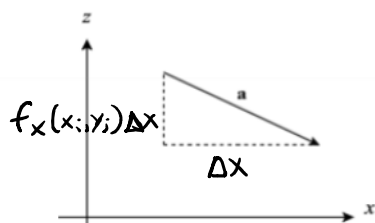
$$\Delta T_{ij} = \|\vec{a} \times \vec{b}\|$$

To complete this computation, we need to find the components of $\mathbf{a}$ and $\mathbf{b}$. From the picture we see that



$$\vec{a} = \langle \Delta x, 0, z_a \rangle \text{ \& } \vec{b} = \langle 0, \Delta y, z_b \rangle$$

To find the z components we note that both $\mathbf{a}$ and $\mathbf{b}$ share the same direction (3D “slope”) as the tangent lines through the point P_{ij} . Based on our knowledge from Chapter 3, we know these tangent lines have slopes of $f_x(x_i, y_j)$ & $f_y(x_i, y_j)$.



and therefore $f_x(x_i, y_j) = \frac{z_a}{\Delta x} \rightarrow z_a = f_x(x_i, y_j) \Delta x$ & similarly for $f_y(x_i, y_j)$

Therefore, $\vec{a} = \langle \Delta x, 0, f_x \Delta x \rangle$ & $\vec{b} = \langle 0, \Delta y, f_y \Delta y \rangle$

Now we can compute the area of the parallelogram determined by vectors $\mathbf{a}$ and $\mathbf{b}$.

$$\Delta T_{ij} = \|\vec{a} \times \vec{b}\| = \left\| \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \Delta x & 0 & f_x \Delta x \\ 0 & \Delta y & f_y \Delta y \end{vmatrix} \right\| = \left\| \langle -f_x \Delta x \Delta y, -f_y \Delta x \Delta y, \Delta x \Delta y \rangle \right\|$$

Let $\Delta x \Delta y = \Delta A$

$$= \left\| \langle -f_x \Delta A, -f_y \Delta A, \Delta A \rangle \right\|$$

$$= \sqrt{f_x^2 \Delta A^2 + f_y^2 \Delta A^2 + \Delta A^2}$$

$$= \sqrt{f_x^2 + f_y^2 + 1} \cdot \Delta A$$

So $S = \lim_{n, m \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n \sqrt{f_x^2 + f_y^2 + 1} \Delta A$. This is a double integral!

This, along with a generalization that proves this idea works for surfaces over regions besides rectangles, leads to the following theorem.

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Theorem: The area of the surface with equation $z = f(x, y)$ where $(x, y) \in D$ and the first partials of f are continuous is:

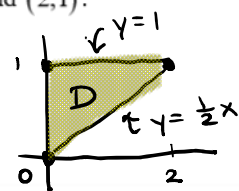
$$S = \iint_D \sqrt{(f_x(x, y))^2 + (f_y(x, y))^2 + 1} dA$$

$$= \iint_D \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dA.$$

Example 1: Set up but do not evaluate an iterated integral which yields the surface area of the surface determined by $z = 1 + 2x + 3y + 4y^2$ that lies above the triangle with vertices $(0, 0)$, $(0, 1)$ and $(2, 1)$.

We first sketch the triangular region D

$$D = \{(x, y) \mid \frac{1}{2}x \leq y \leq 1 \text{ \& } 0 \leq x \leq 2\}$$



$$S = \int_0^2 \int_{\frac{1}{2}x}^1 \sqrt{(2)^2 + (3+8y)^2 + 1} dy dx$$

Example 2: Find the exact surface area of the part of the surface given by $z = 4 - x^2 - y^2$ that lies above the xy -plane.

The intersection between the xy -plane ($z=0$)

& $z = 4 - x^2 - y^2$ is

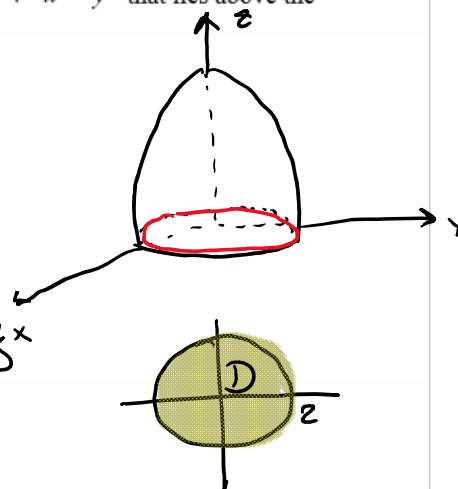
$$0 = 4 - x^2 - y^2 \rightarrow \underline{x^2 + y^2 = 4}$$

We sketch a picture of D

$$D = \{(r, \theta) \mid 0 \leq r \leq 2 \text{ \& } 0 \leq \theta \leq 2\pi\}$$

$$S = \int_0^{2\pi} \int_0^2 \sqrt{(-2x)^2 + (-2y)^2 + 1} \cdot r dr d\theta$$

$$= \int_0^{2\pi} \int_0^2 \sqrt{4r^2 + 1} r dr d\theta$$



Parametric Surfaces and Their Areas

Recall:

In chapter 2 we learned that vector functions of the form $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^3$ is a powerful way to represent curves in the three space. We also learned that vector functions of this form could do a very nice job modeling the movement of a particle in three dimensions.

Then, in chapter 3 we learned that the graph of a function of two variables can be graphed as a surface in three dimensions.

In this section we generalize the ideas presented in chapter 2 to surfaces. This leads to a more powerful way of representing surfaces (than was learned in chapter 3).

Definition: The set of all points in $\mathbb{R}^3$ such that $x = x(u, v)$ $y = y(u, v)$ $z = z(u, v)$ where (u, v) varies throughout some subset, D , of $\mathbb{R}^2$, is called a parametric surface σ and the three equations are called the parametric equations of σ .

Note: Each choice of (u, v) gives a point on σ ; by making all possible choices we get all of σ . A nice illustration of how this works can be found in figure 14.4.5 on page 1028 of the book.

Note: We can represent a parametric surface by a vector function of the form

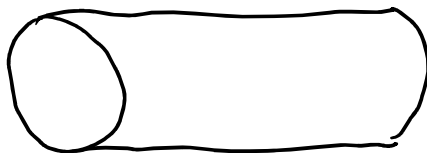
$\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$ where we obtain a picture of the surface by thinking of each output of $\mathbf{r}$ as a position vector and we graph only its terminal point.

Example 3: Identify and sketch the surface given by the vector function

$$\mathbf{r}(u, v) = (2\cos(u))\mathbf{i} + (v)\mathbf{j} + (2\sin(u))\mathbf{k}.$$

We have a circle of radius 2 in xz -directions

The y variable operates independently.



A cylinder.

Example 4: Find parametric equations of the plane given by $x + y + z = 1$.

$$\hookrightarrow z = 1 - x - y$$

$$\text{let } \begin{cases} x = u \\ y = v \end{cases}$$

$$\begin{cases} y = v \\ z = 1 - u - v \end{cases}$$

$$\begin{cases} z = 1 - u - v \end{cases}$$

$$\text{So } \vec{r}(u, v) = \langle u, v, 1 - u - v \rangle$$

Example 5:

a) What type of quadric surface does $\mathbf{r}(u, v) = \langle u, v, v^2 - u^2 \rangle$ yield?

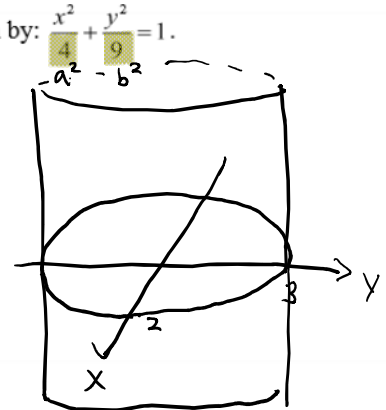
We see $x = u$ & $y = v$ & $z = v^2 - u^2$ or $z = y^2 - x^2$

hyperbolic paraboloid



b) Find a vector valued function $\mathbf{r}(u, v)$ which yields the elliptical cylinder given by: $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

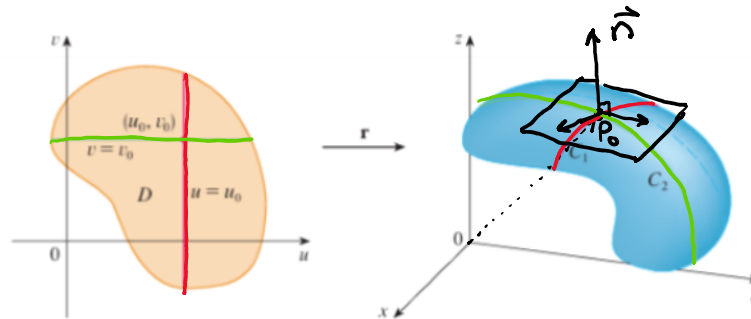
$$\vec{r}(u, v) = \langle 2\cos(u), 3\sin(u), v \rangle$$

**Exploration:**

Now, suppose that we would like to find the tangent plane to the surface σ (at some point on σ) where σ is traced out by a vector function $\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$.

Specifically suppose we want the tangent plane to the surface at some point P_0 with position vector $\mathbf{r}(u_0, v_0)$.

Suppose we consider the grid curve on σ formed by $\mathbf{r}(u, v)$ when we let $u = u_0$ and the grid curve on σ formed by $\mathbf{r}(u, v)$ when we let $v = v_0$. A picture of the situation is:



The curve on σ in red is the grid curve formed by $\vec{r}(u_0, v)$

The curve on σ in green is the grid curve formed by $\vec{r}(u, v_0)$

We want the tangent plane to contain both the vectors $\vec{r}_u(u_0, v_0)$ & $\vec{r}_v(u_0, v_0)$

So let $\mathbf{n} = \vec{r}_u(u_0, v_0) \times \vec{r}_v(u_0, v_0)$

Definition: Suppose a surface σ is given by $\mathbf{r}(u,v) = x(u,v)\mathbf{i} + y(u,v)\mathbf{j} + z(u,v)\mathbf{k}$. If $\mathbf{r}_u \times \mathbf{r}_v \neq \mathbf{0}$ then the surface σ is called smooth.

Note: Intuitively, a surface is smooth if it has no sharp corners. We can also say that a surface, σ , is smooth if we can find a tangent plane to σ at every point on σ .

Fact: Suppose a surface σ is given by $\mathbf{r}(u,v) = x(u,v)\mathbf{i} + y(u,v)\mathbf{j} + z(u,v)\mathbf{k}$. If $\mathbf{r}(u_0, v_0) = \langle x_0, y_0, z_0 \rangle$ and $\mathbf{r}_u(u_0, v_0) \times \mathbf{r}_v(u_0, v_0) \neq \mathbf{0}$, then an equation of the tangent plane to σ at the point (x_0, y_0, z_0) is

$$(\mathbf{r}_u(u_0, v_0) \times \mathbf{r}_v(u_0, v_0)) \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$$

Example 6: Find an equation of the tangent plane to the surface given by $\mathbf{r}(u,v) = \langle u^2, v^2, u+2v \rangle$ at the point $(1,1,3)$.

Note $\mathbf{r}(u,v)$ arrives at $(1,1,3)$ when $u=1$ & $v=1 \Rightarrow (1,1)$

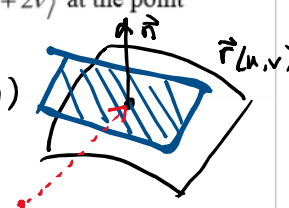
$$\mathbf{r}_u = \langle 2u, 0, 1 \rangle \rightarrow \text{at } (1,1) \quad \mathbf{r}_u(1,1) = \langle 2, 0, 1 \rangle$$

$$\mathbf{r}_v = \langle 0, 2v, 2 \rangle \rightarrow \text{at } (1,1) \quad \mathbf{r}_v(1,1) = \langle 0, 2, 2 \rangle$$

$$\text{So } \mathbf{r}_u(1,1) \times \mathbf{r}_v(1,1) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 1 \\ 0 & 2 & 2 \end{vmatrix} = \langle -2, -4, 4 \rangle = \hat{n}$$

Equation of tangent plane is

$$-2(x-1) - 4(y-1) + 4(z-3) = 0$$



As we did previously we can approximate the surface area of a surface by using portions of tangent planes, and as our partition gets finer our approximation approaches perfection.

This idea leads to the following definition.

Definition: If a smooth parametric surface σ is given by the equation $\mathbf{r}(u,v) = x(u,v)\mathbf{i} + y(u,v)\mathbf{j} + z(u,v)\mathbf{k}$ where $(u,v) \in D$ and σ is covered just once as (u,v) ranges throughout the parameter domain D , then the surface area of σ is $S = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\| dA$.

Note: See page 1034-1035 of the book for the Riemann sum version of the above definition.

Example 7: Find the exact surface area of the portion of the spiral ramp given by $\mathbf{r}(u,v) = \langle u \cos(v), u \sin(v), v \rangle$ where $(u,v) \in \{(u,v) : 0 \leq u \leq 1, 0 \leq v \leq u\}$

Note $\vec{r}_u = \langle \cos(v), \sin(v), 0 \rangle$ ↖ Region D is given!

$$\vec{r}_v = \langle -u \sin(v), u \cos(v), 1 \rangle$$

$$\text{So } \vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \cos(v) & \sin(v) & 0 \\ -u \sin(v) & u \cos(v) & 1 \end{vmatrix} = \langle \sin(v), -\cos(v), u \cos^2(v) + u \sin^2(v) \rangle$$

$$= \langle \sin(v), -\cos(v), u \rangle$$

$$\text{So } \|\vec{r}_u \times \vec{r}_v\| = \sqrt{\sin^2(v) + \cos^2(v) + u^2} = \sqrt{1+u^2}$$

$$\text{Thus, } S = \int_0^1 \int_0^u \sqrt{1+u^2} \, dv \, du$$

Remark: Suppose we want to find the surface area of the surface determined by $z = f(x,y)$ where $(x,y) \in D$. Rather than using the ideas from the beginning of this section, we can do this with the theory developed about parametric surfaces.

We first parameterize $z = f(x,y)$ by letting $x = u$, $y = v$, & $z = f(u,v)$

$$\text{So } \vec{r}(u,v) = \langle u, v, f(u,v) \rangle$$

$$\text{Clearly } \vec{r}_u = \langle 1, 0, f_u(u,v) \rangle \text{ \& } \vec{r}_v = \langle 0, 1, f_v(u,v) \rangle$$

$$\vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & f_u \\ 0 & 1 & f_v \end{vmatrix} = \langle -f_u, -f_v, 1 \rangle$$

$$\|\vec{r}_u \times \vec{r}_v\| = \sqrt{f_u^2 + f_v^2 + 1}$$

$$\text{So } S = \iint_{D_{uv}} \sqrt{f_u^2 + f_v^2 + 1} \, dA \quad \text{Since } x = u \text{ \& } y = v$$

$$\text{then } S = \iint_{D_{xy}} \sqrt{f_x^2 + f_y^2 + 1} \, dA$$